

Figure 4. The eq. 17 inhibitor source clause across the lineage

Paper Eq. 17 (Salazar-Ciudad and Jernvall 2010, p. 586)

$$\partial[\text{Inh}]/\partial t = [\text{Act}] - k_{deg}[\text{Inh}] + k_{di} \nabla^2[\text{Inh}]$$

(applies when knot OR $d_i \geq k_{int}$)

Source: activator concentration

*FORTRAN translation
(d_i ramp added; cosmetic)*

13.f90:487 (humppa_translate.f90:617 byte-identical)

```
hq3d(i,1,2) = q3d(i,1,1)*DiffState(i)
              - mu * q3d(i,1,2)
```

Source: [Act] x d_i (concentration; smooth d_i ramp added)

*Python translation
*** silent variable substitution ***
q3d(i,1,1) -> hq3d[i,0,0]*

coreop2d.py:647 (Path A Python translation)

```
hq3d[i, 0, 1] = hq3d[i, 0, 0] * cls.diff_state[i] \
                - cls.Deg * cls.q3d[i, 0, 1]
```

Source: (rate of [Act]) x d_i -- rate variable, NOT concentration

*configurable comparative
(captured as three named options)*

silicoshark v2 Discretisation.eq17_inh_source (THIS WORK)

```
'act_concentration' -> PAPER_2010 / PATH_B_DEFAULT
'act_times_di'      -> LEGACY_FORTRAN / HUMPPA_LITERAL
'act_rate_times_di' -> PATH_A_REWRITE
```

All three forms named, cited, and exercisable in the comparison study